



Exploratory Data Analysis

Assessment 1 · Group030

FIT5196 Data Wrangling

King Man Chan · Yinglin Fang · Sizhe Hong

Xinhang Ren · Guohou Zhang

1. Data and analytical approach

1.1 Start from the submitted relational data

The analysis begins by loading the six standardised CSV files produced in the solution notebook. The tables contain 5,000 orders, 15,723 order-item lines, 500 customers, 5,000 deliveries, 1,000 products and 7,000 reviews. Identifiers remain strings; only the numeric, date and Boolean fields used below are converted after loading.

Table	Rows	Columns	Role in the EDA
orders	5,000	23	Order value, discount and time
order_items	15,723	6	Product-level sales contribution
customers	500	20	Prior customer activity
deliveries	5,000	20	OTIF and fulfilment
products	1,000	21	Category and catalogue cost
product_reviews	7,000	21	Rating and review timing

1.2 Check the data before analysis

Before choosing the figures, we rechecked that the row count of each table matched its primary-key count. All six checks passed. For relational figures, the parent key was required to be unique and the left-hand row count had to remain unchanged. Orders were aggregated before the customer analysis, and multiple reviews were averaged within order before the delivery join. These checks protect the observation unit used in each chart.

The solution notebook provides the complete transformation and validation record. For this EDA, the most relevant results are zero orphan keys across the submitted relationships, a \$0.00 maximum arithmetic difference, consistent delivery and OTIF fields, and preservation of the 303 reviews containing non-Latin script.

1.3 Move from table structure to analytical questions

We first examined the distribution of order value, then compared discounts, product categories and months. We next moved to customer behaviour, operational performance and the relationship between delivery timing and reviews. This sequence progresses from single-table description to checked relational analysis.

Continuous group means are reported with 95% confidence intervals, and OTIF proportions use Wilson 95% intervals. The figures describe associations in this export and are not interpreted as causal effects.

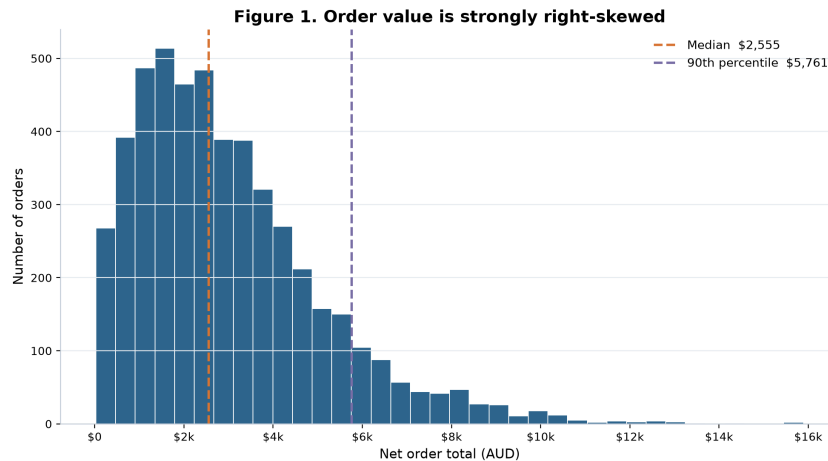
1.4 Assessed EDA coverage

Requirement	Evidence	Observation unit
Univariate distribution	Figure 1	Order
Bivariate comparison	Figure 2	Order within discount group
Multivariate analysis	Figure 3	Order-item line
Temporal pattern	Figure 4	Calendar month
Review behaviour	Figure 7	Rated order
Delivery performance	Figure 6	Delivery
Checked relational analysis	Figures 3, 5 and 7	Item, customer and rated order

2. Assessed visualisations

2.1 What does a typical order look like, and where does the high-value tail begin?

Method and rationale. Following the distribution-first approach described by NIST/SEMATECH (n.d.), a histogram retains the shape of the full order-value distribution. The median and empirical 90th percentile were added because the right-skewed distribution makes the mean alone a poor description of a typical order.

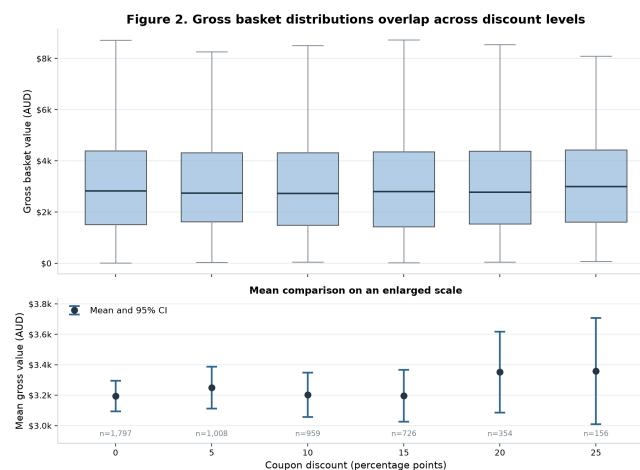


Unit and denominator: one order; all 5,000 canonical orders. **Table:** orders; no join. **Measure:** order_total, with the median and empirical 90th percentile.

The median order is **\$2,555.44**, while the highest 10% begin at **\$5,760.76**. The long upper tail means that a mean alone would overstate a typical order. This is a value threshold rather than a profitability threshold because costs and discounts are not allocated at order level.

2.2 Are deeper discounts associated with larger gross baskets?

Method and rationale. Boxplots compare the median, middle 50% and overall spread without displaying thousands of overlapping points. A separate enlarged panel reports group means and 95% confidence intervals so uncertainty remains visible even though the full distributions are wide.



Unit and denominator: one order within each coupon-discount group; all 5,000 orders. **Table:** orders; no join. **Measure:** gross_order_price before discount, shown by median, interquartile range and mean.

Mean gross basket value ranges only from about **\$3,195 to \$3,359**, and the interquartile ranges overlap heavily. The 20% and 25% groups are much smaller, so their mean estimates are less precise. This is an observational comparison: coupon targeting and customer/product mix may differ between groups.

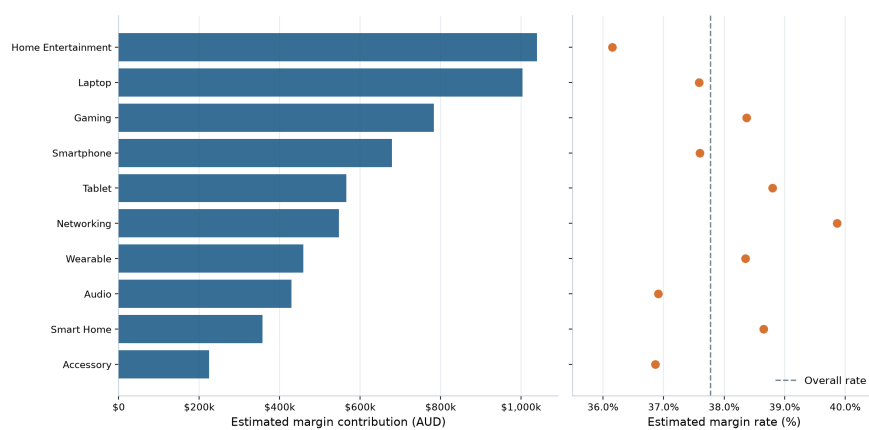
Table 1. Discount-group summary

Discount	Orders	Mean gross basket	Median	Mean 95% CI
0%	1,797	\$3,194.58	\$2,830.16	±\$100.69
5%	1,008	\$3,249.81	\$2,741.96	±\$137.06
10%	959	\$3,202.16	\$2,731.51	±\$146.45
15%	726	\$3,195.56	\$2,807.36	±\$170.41
20%	354	\$3,351.24	\$2,780.20	±\$265.50
25%	156	\$3,358.61	\$3,004.11	±\$348.44

2.3 Which product categories combine economic scale with a strong estimated margin rate?

Method and rationale. The two aligned panels use a common category order. Horizontal bars show total estimated contribution, while points show margin rate; separating these measures prevents a high-volume category from being mistaken for a high-rate category.

Figure 3. Category scale and estimated margin rate tell different stories



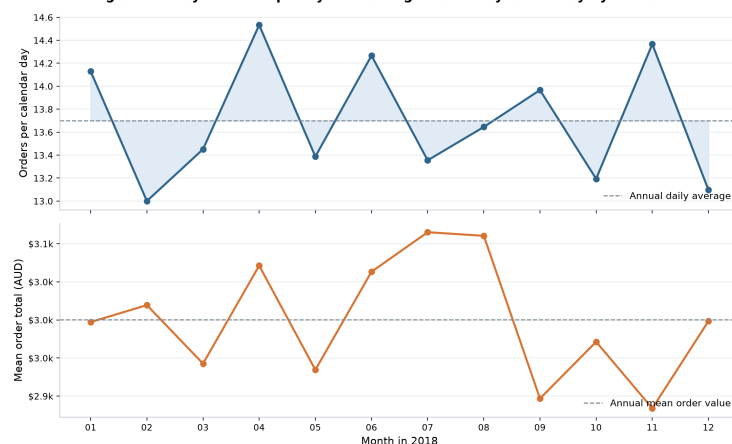
Unit and denominator: one canonical order-item line; all 15,723 lines. **Tables and join:** `order_items.product_id` → `products.product_id`, many-to-one. **Measures:** estimated margin contribution and estimated margin rate by category.

Home Entertainment and Laptop contribute the largest estimated margin dollars, while Networking has the highest estimated rate. These values use catalogue unit cost and do not allocate coupon discounts, returns or overhead, so they should be described as an exploratory contribution measure rather than accounting profit.

2.4 Do monthly changes come from order frequency or order value?

Method and rationale. Orders are aggregated by calendar month and divided by the number of days in that month. The coordinated panels then compare transaction frequency with mean order value without treating a longer month as stronger demand.

Figure 4. Daily order frequency and average value vary differently by month



Unit and denominator: calendar month, calculated from all orders in that month. **Table:** `orders`; no join. **Measures:** orders per calendar day and mean net order value.

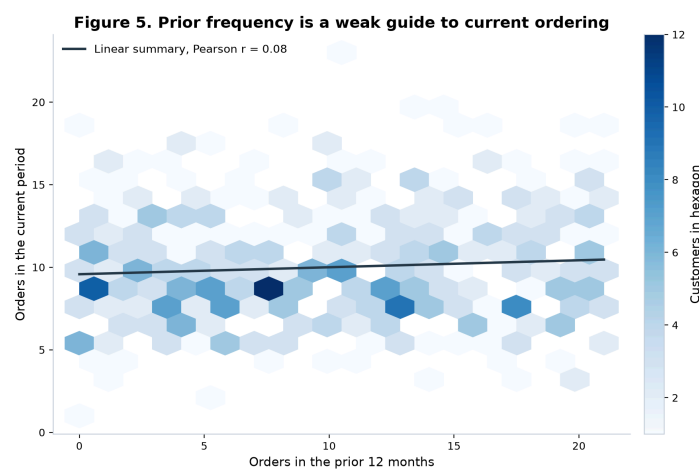
Daily order frequency and average order value move differently rather than following one common trend. With only one calendar year and no campaign calendar, the pattern is useful for description but not enough to claim seasonality.

Table 2. Monthly values used in Figure 4

Month	Orders/day	Mean order value	Month	Orders/day	Mean order value
Jan	14.13	\$2,997.25	Jul	13.35	\$3,115.23
Feb	13.00	\$3,019.50	Aug	13.65	\$3,110.49
Mar	13.45	\$2,942.67	Sep	13.97	\$2,896.78
Apr	14.53	\$3,071.36	Oct	13.19	\$2,971.34
May	13.39	\$2,934.45	Nov	14.37	\$2,884.01
Jun	14.27	\$3,063.32	Dec	13.10	\$2,998.45

2.5 Does a customer's prior order frequency carry into the current period?

Method and rationale. Both variables are integer customer counts, so a conventional scatter plot would hide repeated observations. Hexagonal bins show how many customers occupy each region, and the fitted line is included only as a summary of the weak linear association.

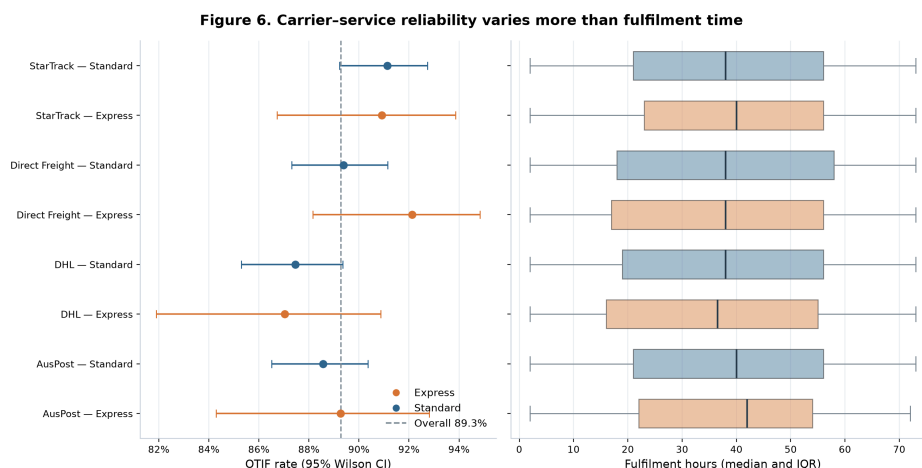


Unit and denominator: one customer; all 500 customers. **Tables and join:** current orders are first aggregated to customer grain, then joined to `customers.customer_id` one-to-one. **Measures:** prior 12-month orders and current-period order count.

The correlation is only **0.08** across 500 customers. Historical frequency alone therefore provides little separation in this period. This does not mean customer history is useless: tenure, marketing exposure and product mix may explain changes that a two-variable plot cannot show.

2.6 Which carrier and service combinations differ in reliability or fulfilment time?

Method and rationale. OTIF is a binary proportion, so point estimates are shown with Wilson 95% intervals. Fulfilment hours are shown as boxplots because their within-group distributions overlap and cannot be represented adequately by one mean.



Unit and denominator: one completed delivery within each carrier-service group; all 5,000 deliveries. **Tables and join:** the submitted delivery table already carries carrier, service, OTIF and fulfilment fields; its `order_id` relationship to orders was checked in the solution workflow. No extra metric-producing join is required here. **Measures:** OTIF rate with Wilson 95% CI, and fulfilment-hours distribution.

Overall OTIF is **89.3%**. The group estimates differ, but several intervals overlap, so the chart does not support a simple best-to-worst league table. Fulfilment-hour distributions also overlap substantially. Route mix, warehouse workload and destination distance remain plausible alternative

explanations.

Table 3. Delivery-group estimates

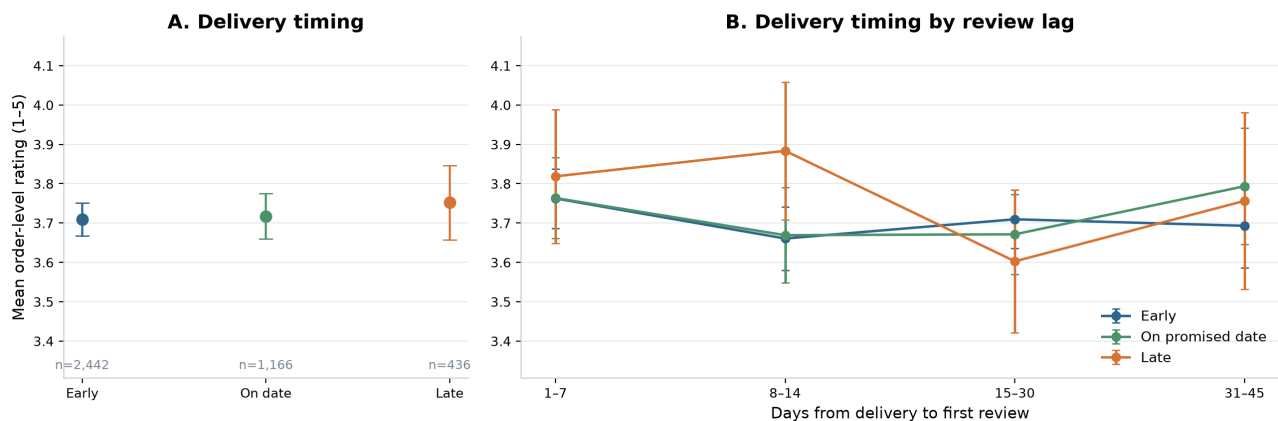
Carrier	Service	n	OTIF (95% Wilson CI)	Median fulfilment
AusPost	Express	205	89.3% (84.3–92.8%)	42.0 h
AusPost	Standard	1,050	88.6% (86.5–90.4%)	40.0 h
DHL	Express	216	87.0% (81.9–90.9%)	36.5 h
DHL	Standard	1,029	87.5% (85.3–89.4%)	38.0 h
Direct Freight	Express	254	92.1% (88.2–94.9%)	38.0 h
Direct Freight	Standard	989	89.4% (87.3–91.2%)	38.0 h
StarTrack	Express	253	90.9% (86.7–93.9%)	40.0 h
StarTrack	Standard	1,004	91.1% (89.2–92.7%)	38.0 h

2.7 Are delivery timing and review delay associated with the rating left for an order?

Method and rationale. The delivery- and review-timing comparison was motivated by Ravula (2023). Reviews are first averaged within order, preventing orders with several reviews from receiving extra weight. Point estimates and 95% intervals compare groups; the enlarged vertical scale is labelled explicitly because all means occupy a narrow part of the original 1–5 scale.

Figure 7. Rating differences are small across delivery and review timing

Enlarged vertical scale; all values remain within the original 1–5 rating scale.



Unit and denominator: one order with at least one review; 4,044 rated orders. Multiple reviews are averaged within order before joining.

Tables and join: order-level `product_reviews.order_id` → `deliveries.order_id`, one-to-one after aggregation. **Measures:** mean order-level rating by early/on-date/late delivery and by days from delivery to first review, with 95% CIs on the full 1–5 scale.

Mean order-level ratings are **3.71 for early**, **3.72 for on-date** and **3.75 for late** deliveries. The confidence intervals overlap, and the review-lag panel has no stable monotonic pattern. This differs from the simple expectation that late delivery must produce a lower mean rating. Reviewer selection, product experience and the small late-delivery groups within each lag band could mask an association; this descriptive figure is not a causal test.

Table 4. Order-level rating by delivery timing

Delivery timing	Rated orders	Mean rating	Mean 95% CI
Early	2,442	3.709	3.667–3.751
On promised date	1,166	3.717	3.659–3.775
Late	436	3.752	3.658–3.846

3. Evidence-based findings

- 3.1 Order value is concentrated in a long upper tail (Figure 1).** Across all 5,000 canonical orders, the median net total is **\$2,555.44**, while the highest 10% begin at **\$5,760.76**. The tail matters because a small group of orders accounts for unusually high transaction value. Product mix and quantity may explain these orders, so the threshold is suitable for investigation rather than automatic customer treatment.
- 3.2 Deeper discounts are not accompanied by a clear increase in gross basket size (Figure 2).** Mean gross order value ranges from **\$3,194.58 to \$3,358.61** across the six discount groups, and their distributions overlap substantially. The 20% and 25% groups contain only 354 and 156 orders, so their confidence intervals are wider. Coupon allocation is observational; a controlled campaign would be needed before concluding that discount depth changes spending.
- 3.3 Home Entertainment and Laptop provide the largest estimated margin pools (Figure 3).** At order-item grain, Home Entertainment contributes approximately **\$1.040 million** and Laptop **\$1.004 million** across 15,723 lines. This suggests that changes in these categories could have a large aggregate effect. The calculation uses catalogue cost and excludes allocated coupons, returns and overhead, so it is an operating indicator rather than accounting profit.
- 3.4 The category with the highest estimated margin rate is not the largest contributor (Figure 3).** Networking has the highest estimated rate at **39.9%**, but its estimated contribution is about **\$547,039**, roughly half that of Home Entertainment. Rate and scale therefore lead to different category priorities. Before changing assortment, realised discounts and category-specific operating costs should be included.
- 3.5 Monthly order frequency and average value do not move together (Figure 4).** Daily order frequency ranges from **13.0 orders in February to 14.53 in April**, while mean order value ranges from **\$2,884.01 in November to \$3,115.23 in July**. Capacity and revenue planning should therefore treat transaction volume and basket value separately. One year of data and no campaign calendar are insufficient to identify a stable seasonal pattern.
- 3.6 Prior order frequency is a weak stand-alone guide to current activity (Figure 5).** Across all 500 customers, prior-12-month orders and current-period orders have a Pearson correlation of only **0.08**. A retention rule based only on previous frequency would provide limited separation. Tenure, campaign exposure and product preference may account for changes, so these factors should be tested before using the relationship operationally.
- 3.7 Most deliveries meet the OTIF definition, but the remaining failure share is still material (Figure 6).** Of 5,000 deliveries, **89.3%** are on time and in full, leaving approximately one in nine outside the target. This provides a clear operational outcome for further investigation. The aggregate rate hides differences in route difficulty, service mix and warehouse conditions, so interventions should not be assigned from the overall figure alone.
- 3.8 Observed carrier–service OTIF rates differ, but several uncertainty intervals overlap (Figure 6).** DHL Express has the lowest observed rate at **87.0% (n=216)** and Direct Freight Express the highest at **92.1% (n=254)**. The overlap in Wilson intervals means the sample does not support a definitive league table. A fair comparison would adjust for distance, warehouse, order mix and repeated performance over time.
- 3.9 Express service does not consistently produce shorter fulfilment inside the warehouse (Figure 6).** Median fulfilment ranges from **36.5 to 42.0 hours** across carrier–service groups, and the distributions overlap broadly. This suggests that dispatch preparation and external transport speed should be studied separately. Carrier selection may also be related to route and order characteristics, which are alternative explanations for the observed differences.

3.10 Average rating shows no clear disadvantage for late delivery in this sample (Figure 7). Among 4,044 rated orders, mean ratings are **3.71 for early (n=2,442)**, **3.72 for on-date (n=1,166)** and **3.75 for late delivery (n=436)**, with overlapping confidence intervals. This does not show that delays are harmless: only reviewed orders are observed, and product experience may dominate the average. Low-rating risk and delay severity deserve separate analysis.

4. Future machine-learning questions

No model is trained in this assignment. Each question is grounded in an EDA result and uses predictors available at the proposed decision time.

4.1 OTIF-failure classification

Can we identify orders at risk of OTIF failure early enough to prioritise intervention? Figure 6 shows a 10.7% overall failure share and uncertain carrier–service differences.

Analysis design. One order; binary target `on_time_in_full=False`. Candidate predictors are Warehouse, order time, basket value and size, expedited flag, carrier, service level and shipping distance, using only fields known by dispatch.

Evaluation and risk. Forward time split; PR-AUC, calibration and recall at the intervention capacity. Delivered date, delay reason and completed tracking events would leak the outcome; compare errors across locations and service levels.

4.2 Fulfilment-hours regression

Can fulfilment hours be predicted when warehouse work begins so staffing or escalation can be planned? Figure 6 shows broad and overlapping fulfilment distributions.

Analysis design. One delivery/order; regression target `fulfilment_hours`. Candidate predictors are Order timestamp, warehouse, item count, total quantity, basket value, expedited flag, service level and distance available at fulfilment start.

Evaluation and risk. Rolling temporal split; MAE and 90th-percentile absolute error. Dispatch/delivery timestamps and later tracking events leak the target; monitor systematic underprediction by warehouse and service.

4.3 Low-rating classification

Which completed orders are at risk of receiving a 1–2 star review so service recovery can be prioritised? Figure 7 shows that simple delivery timing means do not separate ratings well.

Analysis design. One completed order that is eligible for review; binary low-rating target after aggregating multiple reviews consistently. Candidate predictors are Product/category mix, customer history and delivery facts available before outreach.

Evaluation and risk. Time-based split by order completion; PR-AUC, recall at contact capacity and calibration. Review text, helpful votes and realised rating fields leak the outcome; non-reviewers create selection bias and performance should be checked by language.

4.4 Future customer-frequency regression

Can the number of orders a customer will place in the next 90 days support retention planning? Figure 5 shows prior frequency alone has weak association.

Analysis design. One customer at a fixed prediction date; count target of orders in the following 90 days. Candidate predictors are Orders strictly before the prediction date, recency/frequency/value summaries, tenure, tier, acquisition source and previous category mix.

Evaluation and risk. Successive cohort-based temporal splits; MAE and Poisson deviance. Future orders and retrospectively updated lifetime value leak the target; customer attributes may proxy sensitive characteristics.

4.5 Product clustering

Can products be grouped into stable commercial roles for assortment review? Figure 3 shows that contribution and margin rate describe different category positions.

Analysis design. One product; unsupervised clusters based on lagged demand, revenue, estimated margin, price and review summaries. Candidate predictors are Only measures calculated from a completed historical window, standardised before clustering.

Evaluation and risk. Compare stability across adjacent time windows; silhouette score plus merchant interpretability. Scale choices can dominate clusters, sparse products may be unstable, and catalogue cost does not represent full realised profitability.

5. Limitations and conclusion

This analysis covers one retailer and one calendar year. It does not include campaign schedules, returns, realised operating costs, warehouse workload or route difficulty. Coupon, carrier and service assignments were not random, and product reviews represent customers who chose to leave feedback. The figures therefore describe associations in this export rather than causal effects. Estimated category margin should not be read as accounting profit, and the monthly analysis is too short to establish seasonality.

The strongest descriptive signals are the concentration of estimated category contribution and the remaining OTIF failure share. By contrast, discount depth does not clearly separate gross basket size, prior frequency is weak on its own, and average ratings differ little by delivery timing. These weak relationships are still useful because they show where simple rules would be unreliable. The next stage should use time-based evaluation, prevent outcome leakage and report uncertainty and subgroup performance before any operational use.

References

NIST/SEMATECH. (n.d.). *Exploratory data analysis techniques*. NIST/SEMATECH e-Handbook of Statistical Methods. <https://www.itl.nist.gov/div898/handbook/eda/section3/eda3.htm>

Ravula, P. (2023). Impact of delivery performance on online review ratings: The role of temporal distance of ratings. *Journal of Marketing Analytics*, 11(2), 149–158. <https://doi.org/10.1057/s41270-022-00168-5>

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The external study is used as motivation only. We do not reproduce its causal model or treat its reported effects as Group030 results.